

# Bit-Parallel Operations to Investigate Properties of Logical Vectors by Logical Operations

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## Abstract

In this paper, we discuss five problems which can be solved by using bit-parallel approach in constant time. And, these five problems often occur in many problems such as exact string matching problems and approximation string matching problems. According to the approaches for these five problems, we can solve exact string matching problems and approximation string matching problems more efficiently.

## 1 Introduction

There are many papers which discuss how to use bit-parallel techniques to solve string matching algorithms [1, 2, 3, 4, 5, 6, 7, 8]. In these papers, we often encounter a problem in which we have to examine whether a vector consists a special property likes all 0's, all 1's or a consecutive sequence of 1's. In this paper, we shall point out that we have a systematical way to solve such problems. A general definition of our problems is: Given a vector  $A$  of 1's and 0's, find a vector  $C$  such that the value of  $C(i)$  indicates some property of vector  $A$ . Here, 1 and 0 correspond to true and false in logic. It is stipulated that the vector  $C$  can be obtained only by logical operations, such as *and*, *or*, etc. The first problem we shall introduce is the Single For All Problem.

## 2 Single For All Problem

We define the problem in the following:

### Problem 1: The Single For All Problem.

Given a vector  $A = (a_1, a_2, \dots, a_n)$  and  $j$ , find a vector  $C$  such that  $C(k) = 1$  if and only if  $A(j) = A(j+1) = \dots = A(k) = 1$ .

For instance,  $A = (0, 0, 1, 1, 1, 0, 1, 1)$  and  $j = 3$ .  
 $C = (0, 0, 1, 1, 1, 0, 0, 0)$ .

We shall denote Vector  $A$  as the main vector of the problem. To solve Problem 1, we define an auxiliary vector, namely Vector  $B = (0, 0, \dots, b_j = 1, 0, \dots, 0)$  for Vector  $A$ . Then, we can use the following formula to solve Problem 1. Here, the operation  $+$  is the binary addition from left to right.

### Equation 1:

$$C \equiv (((B + A) \text{ xor } A) \text{ and } A)$$

For instance,  $A = (0, 0, 1, 1, 0, 1, 1)$  and  $j = 3$ .  
 $C = (((B + A) \text{ xor } A) \text{ and } A)$

$$\begin{aligned} &= (((0, 0, 1, 0, 0, 0, 0) + (0, 0, 1, 1, 0, 1, 1) \text{ xor } (0, 0, 1, 1, 0, 1, 1)) \\ &\text{ and } (0, 0, 1, 1, 0, 1, 1)) \\ &= ((0, 0, 0, 0, 1, 1, 1) \text{ xor } (0, 0, 1, 1, 0, 1, 1)) \text{ and } (0, 0, 1, 1, 0, 1, 1) \\ &= (0, 0, 1, 1, 1, 0, 0) \text{ and } (0, 0, 1, 1, 0, 1, 1) \\ &= (0, 0, 1, 1, 0, 0, 0) \end{aligned}$$

Since  $C(k) = 1$  for  $k = 3$  to  $k = 4$ , we conclude that  $A(k) = 1$  for  $k = 3$  to  $k = 4$ . Since  $C(7) = 0$ , we conclude that it is not true that  $A(k) = 1$  for  $k = 3$  to  $k = 7$ .

In the following, we shall prove the correctness of  $C \equiv (((B + A) \text{ xor } A) \text{ and } A)$  for Problem 1. That is, we need to prove  $A(i) = 1$  for  $j \leq i \leq k$   
 $\leftrightarrow (((B + A) \text{ xor } A) \text{ and } A)(k) = 1$ .

First we prove Lemma 1 as follows:

### Lemma 1

$$A(i) = 1 \text{ for } j \leq i \leq k$$

$$\rightarrow (((B + A) \text{ xor } A) \text{ and } A)(k) = 1.$$

Since  $B(j) = 1$  and  $A(i) = 1$  for  $j \leq i \leq k$  as assumed, we have  $(B + A)(i) = 0$  for  $j \leq i \leq k$ . Since  $A(k) = 1$  as assumed,  $((B + A) \text{ xor } A)(k) = 1$ . Thus,  $((B + A) \text{ xor } A) \text{ and } A)(k) = 1$ .

Next, we prove lemma 2 as follows:

### Lemma 2

$$(((B + A) \text{ xor } A) \text{ and } A)(k) = 1 \rightarrow A(i) = 1 \text{ for } j \leq i \leq k.$$

We prove by contradiction. Suppose that there exists an  $i$ ,  $j \leq i \leq k$ , such that  $A(i) = 0$ ,  $((B + A) \text{ xor } A) \text{ and } A)(k) = 0$ . There are two cases:

**Case 1:**  $A(k) = 0$ .

In this case, it is obvious that  $((B + A) \text{ xor } A) \text{ and } A)(k) = 0$ .

**Case 2:**  $A(k) = 1$

Without losing generality, we assume that  $l$  is the smallest  $l$ ,  $j \leq l \leq k$  and  $A(l) = 0$ . Since  $B(j) = 1$  and  $A(l) = 0$  as assumed, we have  $(B + A)(l) = 1$  and  $(B + A)(i) = A(i)$  for  $l < i \leq k$ . Thus  $((B + A) \text{ xor } A)(k) = 0$  and therefore  $((B + A) \text{ xor } A) \text{ and } A)(k) = 0$ . Thus we have proved that  $((B + A) \text{ xor } A) \text{ and } A)(k) = 0$  for both cases.

## 3 Single There Exists Problem

In this section, we discuss the Single There Exists problem defined in the following:

### Problem 2: Single There Exists Problem

Given a vector  $A = (a_1, a_2, \dots, a_n)$  and  $j$ , find a vector  $C$  such that  $C(k) = 1$  if and only if  $\exists i(A(i) = 1)$  for  $j \leq i \leq k \leq n$ .

For instance,  $A = (0, 0, 1, 0, 0, 0, 0, 0)$  and  $j = 2$ .  
 $C(3) = C(4) = C(5) = \dots = C(8) = 1$ .

In this problem, we also define an auxiliary vector, namely Vector  $B = (0, 0, \dots, 0, b_j = 1, 1, \dots, 1)$  for the main vector, Vector  $A$ . We use the following equation to solve Problem 2.

### Equation 2

$$C \equiv (((A + B) \text{ xor } B) \text{ or } A)$$

For instance,  $A = (0,0,1,1,0,0,0)$  and  $j=2$ .

$B = (0,1,1,1,1,1,1)$ .

$C = (((A+B) \text{ xor } B) \text{ or } A)$

$= (((0,0,1,1,0,0,0) + (0,1,1,1,1,1,1)) \text{ xor } (0,1,1,1,1,1,1))$

or  $(0,0,1,1,0,0,0)$

$= ((0101000) \text{ xor } (0,1,1,1,1,1,1)) \text{ or } (0,0,1,1,0,0,0)$

$= (0,0,1,0,1,1,1) \text{ or } (0,0,1,1,0,0,0)$

$= (0,0,1,1,1,1,1)$

In the following, we shall prove the correctness of  $C \equiv (((A+B) \text{ xor } B) \text{ or } A)$  for Problem 2. That is, we need to prove Lemma 3 in the following:

**Lemma 3**

$\exists A(i)=1$  for  $j \leq i \leq k$

$\rightarrow (((A+B) \text{ xor } B) \text{ or } A)(k)=1$ .

Without losing generality, we assume  $l$  is the last index such that  $A(l)=1$  where  $j \leq l \leq k$ . There are 2 cases:  $l=k$  and  $j \leq l < k$ . First, we consider  $j \leq l < k$ . Since  $B(i)=0$  for  $1 \leq i < j$ ,  $B(i)=1$  for  $i \geq j$  and  $A(l)=1$  where  $j \leq l < k$ ,  $((A+B) \text{ xor } B)(k)=1$ . Hence,  $(((A+B) \text{ xor } B) \text{ or } A)(k)=1$ . Then, we consider  $l=k$ . Since  $A(k)=1$ ,  $(((A+B) \text{ xor } B) \text{ or } A)(k)=1$ . Table 3.1 illustrates the above reasoning.

**Lemma 4**

$(((A+B) \text{ xor } B) \text{ or } A)(k)=1 \rightarrow \exists A(i)=1$  for  $j \leq i \leq k$ .

We prove by contradiction. We prove that if  $A(i)=0$  for  $j \leq i \leq k$ ,  $(((A+B) \text{ xor } B) \text{ or } A)(k)=0$ . Since  $B(i)=0$  for  $1 \leq i < j$ ,  $B(i)=1$  for  $i \geq j$  and  $A(i)=0$  for  $j \leq i \leq k$ ,  $(A+B)(k)=1$ . Because  $B(k)=1$ ,  $((A+B) \text{ xor } B)(k)=0$ . Because  $A(k)=0$ ,  $(((A+B) \text{ xor } B) \text{ or } A)(k)=0$ .

Let us review the formulas to solve single for all problem and single there exists problem.

For single for all problem, we use Equation 1

$$C \equiv (((B+A) \text{ xor } A) \text{ and } A).$$

For single there exists problem, we use Equation 2

$$C \equiv (((A+B) \text{ xor } B) \text{ or } A).$$

We can see that the last vector is  $A$  for both equations. Other than this, there is a duality property between these two formulas:

$$A \leftrightarrow B$$

$$\text{AND} \leftrightarrow \text{OR}$$

In some sense, when we solve the single for all problem, the auxiliary vector looks like a single there exists vector while when we solve single there exists problem, the auxiliary vector looks

like a single for all vector. We exploit this special property in our later work. We now introduce multiple for all problem.

## 4 Multiple For All Problem

**Problem 3: Multiple For All Problem**

Given  $A = (a_1, a_2, \dots, a_n)$  and  $j_1, j_2, \dots, j_l$  where  $j_i < j_{i+1}$  for  $i = 1$  to  $i = l$ , find Vector  $C$  such that  $C(k_i)=1$  if and only  $A(x)=1$  for  $j_i \leq x \leq k_i$ ,  $i = 1$  to  $i = l$ .

For instance, let  $A = (0,0,1,1,1,0,1,1,0)$ ,  $j_1 = 3$  and  $j_2 = 7$ . Then,  $C = (0,0,1,1,1,0,1,1,0)$ .

For this problem, the auxiliary vector  $B$  is defined as follows:

$B = (0,0, \dots, b_{j_1}=1,0, \dots, 0, b_{j_2}=1,0, \dots, b_{j_l}=1,0, \dots)$

We can solve this problem by using the following equation.

**Equation 3:**

$$C \equiv (((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A$$

For instance,  $A = (0,0,1,1,1,0,1,1,0)$ ,  $j_1 = 3$  and  $j_2 = 7$ . Thus the auxiliary vector is

$$B = (0,0,1,0,0,0,1,0,0)$$

$$C \equiv (((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A$$

$$= (((((0,0,1,0,0,0,1,0,0) \text{ and } (0,0,1,1,1,0,1,1,0)) + (0,0,1,1,1,0,1,1,0)))$$

$$\text{xor } (0,0,1,1,1,0,1,1,0)) \text{ or } (0,0,1,0,0,0,1,0,0) \text{ and } (0,0,1,1,1,0,1,1,0))$$

$$= (((((0,0,1,0,0,0,1,0,0) + (0,0,1,1,1,0,1,1,0)) \text{ xor } (0,0,1,1,1,0,1,1,0)))$$

$$\text{or } (0,0,1,0,0,0,1,0,0) \text{ and } (0,0,1,1,1,0,1,1,0))$$

$$= (((0,0,0,0,0,1,0,0,1) \text{ xor } (0,0,1,1,1,0,1,1,0)) \text{ or } (0,0,1,0,0,0,1,0,0))$$

$$\text{and } (0,0,1,1,1,0,1,1,0))$$

$$= ((0,0,1,1,1,1,1,1,1) \text{ or } (0,0,1,0,0,0,1,0,0))$$

$$\text{and } (0,0,1,1,1,0,1,1,0))$$

$$= ((0,0,1,1,1,1,1,1,0) \text{ and } (0,0,1,1,1,0,1,1,0))$$

$$= (0,0,1,1,1,0,1,1,0)$$

For the auxiliary vector  $B$ , we divide it into  $l$  parts. Let  $B = P_1 P_2 \dots P_l$  where

$P_1 = (0, \dots, b_{j_1}=1, 0, \dots, b_{j_2-1}=0)$  and Part  $i$  be  $P_i = (b_{j_i}=1, 0, \dots, b_{j_{i+1}-1}=0)$  for  $1 < i \leq l$  shown in Figure 1.

$$B = \begin{matrix} P_1 & P_2 & P_3 & \dots & P_l \\ (0,0,\dots,b_{j_1}=1,0,\dots,0) & (b_{j_2}=1,0,\dots,0) & (b_{j_3}=1,0,\dots,0) & \dots & (b_{j_l}=1,0,\dots,0) \end{matrix}$$

Figure 1. Divide auxiliary vector  $B$  into  $l$  parts.

**Lemma 5**

$A(x)=1$  for  $j_i \leq x \leq k_i$ ,  $i = 1$  to  $i = l \rightarrow (((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A(k_i)=1$

In this Lemma, we need to consider two cases:

$P_1$  and  $P_i$  for  $1 < i \leq l$ . First, we consider the case of  $P_1$ . Since  $B(j_1)=1$  and  $A(x)=1$  for

$j_1 \leq x \leq k_1 \leq b_{j_2-1}$  as assumed, we have  $(B \text{ and } A)(x) = B(x)$  for  $j_1 \leq x \leq k_1 \leq b_{j_2-1}$ . By the same assumption, we have

$((B \text{ and } A) + A)(x) = 0$  for  $j_1 \leq x \leq k_1 \leq b_{j_2-1}$ . Since  $A(k_1)=1$  as assumed,

$$(((B \text{ and } A) + A) \text{ xor } A)(k_1) = 1. \quad \text{Thus,}$$

$((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B)(k_i) = 1$  . Since  $A(k_i) = 1$  ,  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 1$  .  
 Next, we consider the case of  $P_i$  where  $1 < i \leq l$  .  
 Let  $D_i = (d_1 = 0, 0, \dots, d_{j_i}, 0, \dots, d_n = 0)$  where  $d_{j_i}$  is the carry bit of  $P_{j_i-1}$  and  $A$  which is 1 or 0.  
 It is obvious that  $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(i) = 1$   
 $\leftrightarrow$   
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B) \text{ and } A)(i) = 1$   
 for  $i \geq j_2$  . We have to prove  $A(x) = 1$  for  $j_i \leq x \leq k_i$  ,  $1 < i \leq l$   
 $\rightarrow ((((B \text{ and } A) + A + D) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 1$  .  
 Here, there are two cases :  $d_{j_i} = 0$  and  $d_{j_i} = 1$  .  
 If  $d_{j_i} = 0$  , it has been proved in Case 1. Hence,  
 we just need to consider the case of  $d_{j_i} = 1$  .  
 Since  $B(j_i) = 1$  and  $A(x) = 1$  for  $j_i \leq x \leq k_i$   
 as assumed, we have  $(B \text{ and } A)(x) = B(x)$  for  
 $j_i \leq x \leq k_i$  . By the same assumption, we have  
 $((B \text{ and } A) + A)(x) = 0$  for  $j_i \leq x \leq k_i$  .  
 By the definition of  $D_i = (d_1 = 0, 0, \dots, d_{j_i}, 0, \dots, d_n = 0)$  where  $d_{j_i}$  is  
 the carry bit of  $P_{j_i-1}$  and  $A$  which is 1 or 0, we  
 have  $((B \text{ and } A) + A + D_i)(j_i) = 1$  and  
 $((B \text{ and } A) + A + D_i)(x) = 0$  for  $j_i + 1 \leq x \leq k_i$  .  
 Since  $A(k_i) = 1$  as assumed,  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B)(k_i) = 1$  .  
 Thus,  $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B)(k_i) = 1$  .  
 Since  $A(k_i) = 1$  ,  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 1$  .

Next, we prove the Lemma 6 as follows:

**Lemma 6**

$((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 1$   
 $\rightarrow A(x) = 1$  for  $j_i \leq x \leq k_i$ ,  $i = 1$  to  $i = l$ .

There are two case:  $P_1$  and  $P_i$  for  $1 < i \leq l$  . First, we consider the case  $P_1$  . We  
 prove by contradiction. We prove that if there  
 exists a  $x$ ,  $j_1 \leq x \leq k_1$  , such that  $A(x) = 0$  ,  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_1) = 0$  .  
 There are two cases:

**Case 1:**  $A(k_1) = 0$  .

In this case, it is obvious that  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_1) = 0$  .

**Case 2:**  $A(k_1) = 1$  .

Without losing generality, we assume that  $x$  is  
 the smallest  $x$ ,  $j_1 \leq x \leq k_1$  and  $A(x) = 0$  .  
 There are two cases:  $A(j_1) = 0$  and  $A(j_1) = 1$  .  
 For  $A(j_1) = 0$  , we have  $(B \text{ and } A)(i) = 0$  for  
 $j_1 \leq i \leq k_1$  . Thus,  $((B \text{ and } A) + A)(i) = A(i)$  for  
 $j_1 \leq i \leq k_1$  . Hence,  
 $((B \text{ and } A) + A) \text{ xor } A)(i) = 0$  for  $j_1 \leq i \leq k_1$  .  
 Therefore,

$((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B)(i) = B(i)$  for  
 $j_1 \leq i \leq k_1$  . Thus,  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(i) = 0$  for  
 $j_1 \leq i \leq k_1$  .

For  $A(j_1) = 1$ , we have  $(B \text{ and } A)(i) = B(i)$  for  
 $j_1 \leq i \leq k_1$  . Since  $B(j_1) = 1$  and  $A(x) = 0$  as  
 assumed, we have  $(B + A)(x) = 1$  and  
 $(B + A)(i) = A(i)$  for  $x < i \leq k_1$  . Thus,

$((B \text{ and } A) + A) \text{ xor } A)(k_1) = 0$  and since  
 $j_1 \leq x \leq k_1$ ,  $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B)(k_1) = 0$  .  
 Thus,  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_1) = 0$  .

Thus, we have proved that  
 $((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 0$  for  
 both cases. Then, we consider  $1 < i \leq l$  . We  
 prove by contradiction. Assume that there  
 exists a  $x$ ,  $j_i \leq x \leq k_i$  , such that  $A(x) = 0$  ,  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 0$  .  
 There are two cases:

**Case 1:**  $A(k_i) = 0$  .

In this case, it is obvious that  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 0$  .

**Case 2:**  $A(k_i) = 1$

Without losing generality, we assume that  $x$  is  
 the smallest  $x$ ,  $j_i \leq x < k_i$  and  $A(x) = 0$  .  
 There are two cases:  $A(j_i) = 0$  and  $A(j_i) = 1$  .  
 For  $A(j_i) = 0$  , we have  $(B \text{ and } A)(y) = 0$  for  
 $j_i \leq y \leq k_i$  . Thus,  $((B \text{ and } A) + A)(y) = A(y)$   
 for  $j_i \leq y \leq k_i$  . There are two cases:  $d_{j_i} = 1$   
 and  $d_{j_i} = 0$  . Since  $d_{j_i} = 0$  , it has been proved  
 by  $P_1$  . Hence, we consider  $d_{j_i} = 1$  .  
 $((B \text{ and } A) + A + D_i)(j_i) = 1$  and  
 $((B \text{ and } A) + A + D_i)(y) = A(y)$  for  $j_i \leq y \leq k_i$  .  
 Since  $A(j_i) = 0$  ,  
 $((B \text{ and } A) + A + D_i)(y) = 0$  for  
 $j_i < y \leq k_i$  . Since  $j_i \leq x < k_i$  ,  $B(k_i) = 0$  ,  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B)(k_i) = 0$  .  
 Thus,  
 $((((B \text{ and } A) + A + D_i) \text{ xor } A) \text{ or } B) \text{ and } A)(k_i) = 0$  .

## 5 Multiple There Exists Problem

**Problem 4: Multiple There Exists Problem**

Given  $A = (a_1, a_2, \dots, a_n)$  and  $j_1, k_1, j_2, k_2, \dots, j_l, k_l$   
 where  $j_i \leq k_i < j_{i+1} \leq k_{i+1}$  for  $i = 1$  to  $i = l$ , find  
 Vector  $C$  such that  $C(k_i) = 1$  if and only  $\exists A(x) = 1$   
 for  $j_i \leq x \leq k_i$ ,  $i = 1$  to  $i = l$ .

Let  $A = (0, 0, 1, 1, 0, 1, 0)$ . If  $j_1 = 2$ ,  $j_2 = 3$   
 and  $j_3 = 5$ . Then  $C = (0, 0, 1, 0, 1, 1, 0)$ .

For this problem, the auxiliary vector  $B$  is defined  
 as follows:

$B = (0, 0, \dots, b_{j_1} = 1, 1, \dots, b_{k_1} = 1, 0, \dots, 0, b_{j_2} = 1, 1, \dots, b_{k_2} = 1, 0, \dots, 0, \dots)$   
 . The vector operation is as follows:

**Equation 4**

$C \equiv (((A \text{ and } B) + B) \text{ xor } B) \text{ or } A$

Given  $A = (0, 0, 1, 1, 0, 1, 0)$ ,  $j_1 = 2$ ,  $k_1 = 3$ ,  
 $j_2 = 5$  and  $k_2 = 7$ . Let  $B = (0, 1, 1, 0, 1, 1, 0)$ .

$$\begin{aligned}
C &= (((((A \text{ and } B) + B) \text{ xor } B) \text{ or } A) \text{ and } B) && \text{The 27th Workshop on Combinatorial Mathematics and Computation Theory} \\
&= ((((((0,0,1,1,1,0,1,0) \text{ and } (0,1,1,0,1,1,1,0)) + (0,1,1,0,1,1,1,0)) \\
&\text{ xor } (0,1,1,0,1,1,1,0)) \text{ or } (0,0,1,1,1,0,1,0)) \text{ and } (0,1,1,0,1,1,1,0)) \\
&= ((((((0,0,1,0,1,0,1,0) + (0,1,1,0,1,1,1,0)) \text{ xor } (0,1,1,0,1,1,1,0)) \\
&\text{ or } (0,0,1,1,1,0,1,0)) \text{ and } (0,1,1,0,1,1,1,0)) \\
&= (((((0,1,0,1,0,0,1,1) \text{ xor } (0,1,1,0,1,1,1,0)) \text{ or } (0,0,1,1,1,0,1,0)) \\
&\text{ and } (0,1,1,0,1,1,1,0)) \\
&= (((0,0,1,1,1,1,0,1) \text{ or } (0,0,1,1,1,0,1,0)) \text{ and } (0,1,1,0,1,1,1,0)) \\
&= ((0,0,1,1,1,1,1,1) \text{ and } (0,1,1,0,1,1,1,0)) \\
&= (0,0,1,0,1,1,1,0) \\
C(k_1) &= C(3) = 1, && \text{because } A(3) = 1. \\
C(k_2) &= C(7) = 1 && \text{because}
\end{aligned}$$

|   |       |   |       |   |   |   |   |   |   |   |
|---|-------|---|-------|---|---|---|---|---|---|---|
|   | $j_1$ |   | $j_2$ |   |   |   |   |   |   |   |
| B | 0     | 1 | 0     | 0 | 1 | 0 | 0 | 1 | 0 | 0 |

If we perform  $(B + A)$  directly, we would get the vector in Table 2.

Table 2 The result of  $(B + A)$  if no modification is done for Problem 3

|                                         |       |   |       |   |   |   |   |   |   |   |
|-----------------------------------------|-------|---|-------|---|---|---|---|---|---|---|
|                                         | $j_1$ |   | $j_2$ |   |   |   |   |   |   |   |
| A                                       | 0     | 1 | 1     | 1 | 0 | 1 | 1 | 1 | 1 | 0 |
| B                                       | 0     | 1 | 0     | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| B+A                                     | 0     | 0 | 0     | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| $(B+A) \text{ xor } A$                  | 0     | 1 | 1     | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| $((B+A) \text{ xor } A) \text{ and } A$ | 0     | 1 | 1     | 1 | 0 | 1 | 1 | 0 | 1 | 0 |

From Table 2, the fact that  $((B + A) \text{ xor } A) \text{ or } B(j_2 + 1) = 1$ , would mistakenly make us conclude that  $A(j_1) = A(j_1 + 1) = \dots = A(j_2) = A(j_2 + 1) = 1$  which is not correct. This is why we need to perform  $(A \text{ and } B)$  to be used as our auxiliary vector as shown in Table 3.

Table 3

|                                                           |       |   |       |   |   |   |   |   |   |   |
|-----------------------------------------------------------|-------|---|-------|---|---|---|---|---|---|---|
|                                                           | $j_1$ |   | $j_2$ |   |   |   |   |   |   |   |
| A                                                         | 0     | 1 | 1     | 1 | 0 | 1 | 1 | 1 | 1 | 0 |
| B                                                         | 0     | 1 | 0     | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| A and B                                                   | 0     | 1 | 0     | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| $(A \text{ and } B) + A$                                  | 0     | 0 | 0     | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| $((A \text{ and } B) + A) \text{ xor } A$                 | 0     | 1 | 1     | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| $((A \text{ and } B) + A) \text{ xor } A) \text{ and } A$ | 0     | 1 | 1     | 1 | 0 | 0 | 0 | 1 | 1 | 0 |

We have another problem to discuss. Let us display the vector formulas for Problems 1 and 3 as follows:

**Problem 1:**  $((B + A) \text{ xor } A) \text{ and } A$

**Problem 3:**  
 $(((((B \text{ and } A) + A) \text{ xor } A) \text{ or } B) \text{ and } A)$

Why do we need an extra  $(\text{or } B)$  operation after  $((B \text{ and } A) + A) \text{ xor } A$  in the formula for Problem 1? Note that the auxiliary vector of problem 1 is  $(0, 0, \dots, b_j = 1, 0, \dots, 0)$  and the auxiliary vector of Problem 3 is  $(0, 0, \dots, b_{j_1}, 0, \dots, b_{j_2} = 1, 0, \dots, 0)$ . If we do not use  $(\text{or } B)$ , we would have the following situation:

|                                           |       |   |       |   |   |   |   |   |   |    |
|-------------------------------------------|-------|---|-------|---|---|---|---|---|---|----|
|                                           | 1     | 2 | 3     | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A                                         | 0     | 0 | 1     | 1 | 1 | 1 | 1 | 0 | 0 | 0  |
| B                                         | 0     | 0 | 0     | 1 | 0 | 1 | 0 | 0 | 0 | 1  |
|                                           | $j_1$ |   | $j_2$ |   |   |   |   |   |   |    |
| A + B                                     | 0     | 0 | 1     | 0 | 0 | 1 | 0 | 1 | 0 | 1  |
| $(A + B) \text{ xor } A$                  | 0     | 0 | 0     | 1 | 1 | 0 | 1 | 1 | 0 | 0  |
| $((A + B) \text{ xor } A) \text{ and } A$ | 0     | 0 | 0     | 1 | 1 | 0 | 1 | 0 | 0 | 0  |

↑  
wrongly indicating  $A(5) = 0 \neq 1$

We can see that an error occurs as we would

think  $A(j_2) = A(5) = 0 \neq 1$ . If  $(\text{or } B)$  is used, we would be able to correct this error as follows:

Table 5

|       |   |       |   |   |   |   |   |   |    |
|-------|---|-------|---|---|---|---|---|---|----|
| 1     | 2 | 3     | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 0     | 0 | 1     | 1 | 1 | 1 | 1 | 0 | 0 | 0  |
| 0     | 0 | 0     | 1 | 0 | 1 | 0 | 0 | 0 | 1  |
| $j_1$ |   | $j_2$ |   |   |   |   |   |   |    |
| 0     | 0 | 1     | 0 | 0 | 1 | 0 | 1 | 0 | 1  |
| 0     | 0 | 0     | 1 | 1 | 0 | 1 | 1 | 0 | 0  |
| 0     | 0 | 0     | 1 | 1 | 1 | 1 | 1 | 0 | 1  |

↑

Recover

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|

$((A + B) \text{ xor } A) \text{ or } B$  and A

Let us now make an interesting observation: When we solve the multiple for all problem, the auxiliary vector  $B$  looks like a multiple there exists case.  $B = (0, 1, 0, \dots, 1, 0, \dots)$ . On the other hand, when we solve the multiple there exist problem, the auxiliary vector  $B$  looks like a multiple for all case.  $B = (0, 1, 1, \dots, 1, 0, 0, \dots, 1, 1, \dots, 1, 0, \dots)$ . Thus there is a duality property between Problem 3 and 4. We can now use this property to solve Problem 5 in the next section.

## 6 Multiple There Exists and For All Problem

### Problem 5: Multiple There Exists and For All Problem

Given  $A = (a_1, a_2, \dots, a_n)$  and  $B = (b_1, b_2, \dots, b_n)$ , find vector  $C$  such that  $C(k) = 1$  if and only if  $\exists j \leq k, A(j) = 1$  and  $\forall i, j \leq i \leq k, B(i) = 1$ .

Let  $A = (0, 1, 0, 0, 0, 1, 0, 1)$  and  $B = (0, 1, 1, 0, 1, 1, 0, 0)$ . Then  $C(4) = 1$  because  $A(2) = 1$  and  $B(2) = B(3) = B(4) = 1$ . Similarly,  $C(7) = 1$  because  $A(7) = 1$  and  $B(7) = 1$ . It can be seen that  $C(6) = 0$ .

To solve Problem 5, we may use vector  $B$  as the auxiliary vector of  $A$  and use vector  $A$  as the auxiliary vector of  $B$ . Since we are solving a Problem 4 on vector  $A$ , we may use  $C \equiv (((((A \text{ and } B) + B) \text{ xor } B) \text{ or } A) \text{ and } B)$  to solve the  $\exists j \leq k, A(j) = 1$  part. We also have to solve the  $\forall i, j \leq i \leq k, B(i) = 1$  part. Thus we have to solve a Problem 3 on vector  $B$ . By using the duality property of vectors for Problems 3 and 4 and the fact that the vectors 1 and 2 are auxiliary to each other, we may only use the following vector operation to solve Problem 5.

### Equation 5

$$C \equiv (((((A \text{ and } B) + B) \text{ xor } B) \text{ or } A) \text{ and } B)$$

Let  $A = (0,1,0,0,0,0,1,0,1)$  and  $B = (0,1,1,1,0,1,1,0,0)$ .  
 $C = (((((A \text{ and } B) + B) \text{ xor } B) \text{ or } A) \text{ and } B)$   
 $= ((((((0,1,0,0,0,0,1,0,1) \text{ and } (0,1,1,1,0,1,1,0,0)) + (0,1,1,1,0,1,1,0,0))$   
 $\text{ xor } (0,1,1,1,0,1,1,0,0)) \text{ or } (0,1,0,0,0,0,1,0,1)) \text{ and } (0,1,1,1,0,1,1,0,0))$   
 $= ((((((0,1,0,0,0,0,1,0,0) + (0,1,1,1,0,1,1,0,0)) \text{ xor } (0,1,1,1,0,1,1,0,0))$   
 $\text{ or } (0,1,0,0,0,0,1,0,1)) \text{ and } (0,1,1,1,0,1,1,0,0))$   
 $= (((((0,0,0,0,1,1,0,1,0) \text{ xor } (0,1,1,1,0,1,1,0,0)) \text{ or } (0,1,0,0,0,0,1,0,1))$   
 $\text{ and } (0,1,1,1,0,1,1,0,0))$   
 $= (((((0,1,1,1,0,1,1,0) \text{ or } (0,1,0,0,0,0,1,0,1)) \text{ and } (0,1,1,1,0,1,1,0,0))$   
 $= ((0,1,1,1,0,1,1,1) \text{ and } (0,1,1,1,0,1,1,0,0))$   
 $= (0,1,1,1,0,1,1,0,0)$   
 $\bar{C}(4) = 1$  because  $A(2) = B(2) = B(3) = B(4) = 1$ .  
 Similarly,  $C(7) = 1$ .  
 But,  $C(6) = 0$ .

## 7 Conclusion

In this paper, we bring up the systematical way to improve many algorithms. It is very effective and useful for exact string matching problem and approximation sting matching problem.

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